AR Pet Horse– By Laiba Siddique

VR Development Platform and Device

- Unity version: 2022.3.42f1
- Mac OS version: Sequila (15.5)
- AR Device: iPad Air (6th Gen; 2024 M2; iPadOS 18.2)

Setup Instructions

To play AR Pet Horse, you must use an Apple device mobile device and MacBook with XCode, preferably both devices having their respective OS versions be versions released the same year. First, download the Unity .zip provided in the box.com folder. Unzip the file, go to Unity, and projects in Unity, and then click add and add project from disk, choosing the unzipped file. Open the project in Unity and connect an Apple mobile device (set to Developer Mode) to your computer. Follow the steps from this tutorial to allow the app to be properly downloaded, build, and run onto your device. After those steps, you can finally start playing.

How to Play

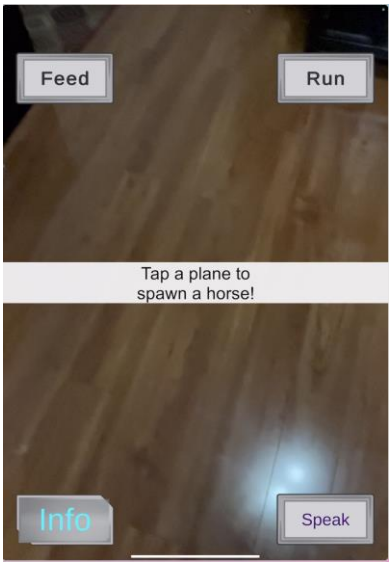


Image 1: Starting UI Instructions and Main UI (First thing that appears when game starts)

Once you have given the app permission to use your camera and your app is running, you are presented with a bar of text explaining that to spawn your AR per horse, you have to tap a plane. To do this, start moving your camera around so it can see and detect horizontal surfaces. Blue planes will start appearing. On a blue plane, tap where you would like to spawn the horse. When the app starts, the main UI will also be visible, but the buttons will not do anything as there is no horse for them to interact with yet.



Image 2: Blue Plane Appearing Before Horse is Spawned In



Image 3: Horse Spawned-In and Horse Energy UI Being Shown

Once the horse is spawned in, you can start interacting with it by pressing the UI buttons. You can press/tap the “feed” button to have the horse eat from its food bowl, which increases its energy. Tapping the “run” button has the horse run for a bit, decreasing its energy. Pressing the “info” button shows/hides the UI presenting the horse’s energy. The horse cannot run when it has no energy <= 0). Feeding the horse gives it 1 energy, and having it run makes it lose 1 energy. You cannot feed the horse while it is running. You cannot have the horse while it is eating either. Lastly, pressing the “speak” button makes the neigh, and hence “speak” (similar to how people command their pet dogs to “bark”).

Known Bugs and Workarounds

- *Not really a bug; moreso an AR quirk I didn’t account for*: There is no proper occlusion culling if you out your hand behind the camera and try to pet the horse that way.